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Analysis and Design of H_∞ Robust Disturbance Observer Based on LMI *

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***Abstract* - Disturbance observer (DOB)-based controller design is one of the most popular methods in the field of motion control. This paper presents a design method for the disturbance observers to satisfy closed loop performance specifications, provided that the main feedback controller is known. Unlike the conventional methods, the proposed DOB design method takes into account the influence from not only disturbance and noise but also model uncertainty to the system's stability and performance. Utilizing the LMI method, the H∞ optimal disturbance observer can be obtained by solving a series of convex optimization problems. Furthermore, by introducing the weighted transfer functions, the proposed method can guarantee the robust stability as well as robust performance of the system.**

Index Terms - disturbance observer (DOB), H_∞ control, LMI.

I. INTRODUCTION

The external disturbance can largely reduce the performance of the motion control system. In order to compensate the external disturbance and improve the performance, K.Ohnishi [1] proposed the concept of disturbance observer. Because DOB is not only able to improve the system's robustness and dynamic performance, but also convenient for application, it's widely used in motion control systems, such as robotic [2], hard disk [3], electric load simulator [12] and etc. The key for the DOB design is to design a low-pass filter $Q(s)$. A binomial form of the Q-filter has been proposed by Umeno.T [4]. Yamada.K [5] has designed a high order DOB to simultaneously satisfy the requirement for fast response and low sensitivity to the disturbance. He additionally pointed out that high order DOB may cause oscillation due to the phase lag. Nevertheless, all of those methods are only qualitative in the Q-filter design, but not in consideration of the influence of model uncertainty.

In consideration of model uncertainty, external disturbance and noise, a novel DOB design method is proposed. The LMI method, which can assure the DOB satisfies the sufficient condition for robust stability of the closed loop, is introduced to this design. Taking advantage of the weighted transfer functions, the DOB performs well in compensating external disturbance. This paper is organized as follows. In section II, the structure and feature of DOB are given. The problem of DOB design for the perturbed system is also described in this section. In sectionⅢ, the DOB design problem is transformed into a H_∞ control problem. The detailed robust DOB design procedure is given in section Ⅳ. Section V gives an example of application, which includes details of the system's construction, feature of the plant and analysis of the simulation.

II. ANALYSIS OF DOB STRUCTURE AND ROBUST DOB DESIGN

In this section, the structure of DOB is first introduced, after which the detailed description of H_∞ robust DOB design problem is presented. By analysing the structure of DOB, it can be seen that DOB design problem is actually a Q-filter design problem. To design a H_∞ robust DOB, we just need to design a proper Q-filter that makes the closed loop system have desired robust performance.

A. The structure of DOB

Affected by the disturbance d and noise n, the control performance is somewhat deteriorated. DOB can estimate and compensate the disturbance from the output of system and controller. The structure of DOB is shown in Fig.1:

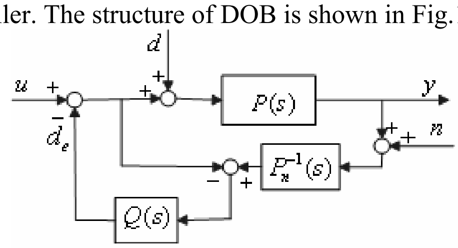

Fig. 1 The structure of DOB

In this figure, $P_n(s)$ is the nominal model of the plant; while $Q(s)$ is a low-pass filter with steady state gain one. From Fig. 1, it can be seen that:

$$y = G_{yu}(s)u + G_{yd}(s)d + G_{yn}(s)n \tag{1}$$

where:

$$G_{yu}(s) = \frac{P(s)P_n(s)}{P_n(s) + Q(s)(P(s) - P_n(s))} \tag{2}$$

$$G_{yd}(s) = \frac{P(s)P_n(s)(1 - Q(s))}{P_n(s) + Q(s)(P(s) - P_n(s))} \tag{3}$$

$$G_{yn}(s) = \frac{P(s)Q(s)}{P_n(s) + Q(s)(P(s) - P_n(s))} \tag{4}$$

Assuming that $P_n(s) \approx P(s)$, since $Q(s) \approx 1$ at low frequencies, it is obviously that $G_{yu}(s) \approx P_n(s)$ and $G_{yd}(s) \approx 0$ from (2) and (3); at high frequencies since $Q(s) \approx 0$, it can be derived from (4) that $G_{yn}(s) \approx 0$. In general, disturbance is at low frequencies, whereas noise is at high frequencies [6]. Therefore, when DOB is added to the system, disturbance can be compensated effectively, and noise almost has no influence on the system output. In a robust internal loop control (RIC) system, the internal loop is comprised of DOB and the plant [6], as is shown in Fig. 3. Because of the compensation of DOB, the input and output of internal loop can approximate the nominal model of the plant. Consequently, the design of controller $C(s)$ can be carried

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out without considering the influence of external disturbance and directly according to the nominal model. Since $G_n^{-1}(s)$ is generally unrealizable, another equivalent form of DOB shown in Fig. 2 is often adopted. In practice, the relative degree of $Q(s)$ should be larger than the relative degree of $G_n(s)$.

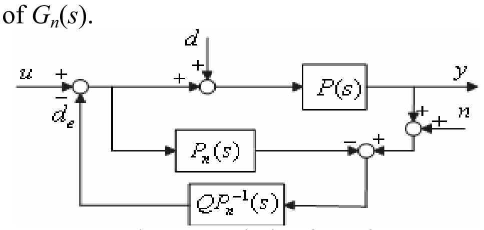


Fig. 2 An equivalent form of DOB

From Fig. 1, it can be seen that in the DOB only $Q(s)$ needs to be solved. If the low-pass $Q(s)$ filter is designed, the problem of designing DOB can be solved. In general, the noise n is high-frequency signal, which can be strengthened by $P_n^{-1}(s)$. Consequently the influence of noise is necessary to be restricted by the low-pass feature of $Q(s)$. Unfortunately, the phase lags of the low-pass filter $Q(s)$ would cause the compensation improperly. Therefore, the design of $Q(s)$ needs a trade-off between the anti-noise ability and disturbance rejection.

B. Analysis of H_∞ robust DOB design

The conventional DOB design methods do not consider directly the model uncertainty. Instead, they treat the uncertainty as an equivalent external disturbance. However, these ways to deal with uncertainty do not take the influence from uncertainty to closed loop into consideration, thus the internal stability can not be guaranteed [13]. The design method proposed in this paper is carried out according to model perturbation. Assuming that some error exists between the practical plant and the nominal model, the modelling error is treated as multiplicative uncertainty, and then we have $P(s) = P_n(s)(1+\Delta_p)$. Provided that controller $C(s)$ has been known, then the RIC system, which is a closed loop system with the disturbance observer and $C(s)$, has the structure as below:

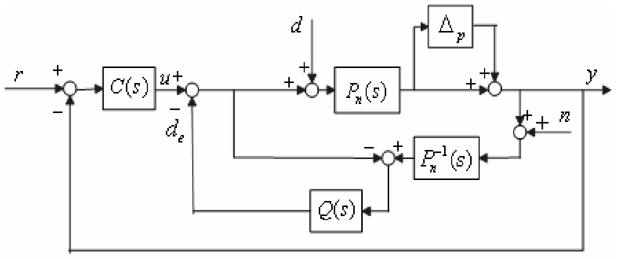


Fig. 3 Robust internal loop control (RIC) system

The design of H_∞ robust DOB should guarantee the system satisfying the following performance requirements:

1) The closed loop system can be kept stable in a given magnitude of uncertainty for the plant.

2) When disturbance, model perturbation and noise act simultaneously, the internal loop of the RIC system can approximate the nominal model well.

Accordingly, the original system is transformed into the following form.

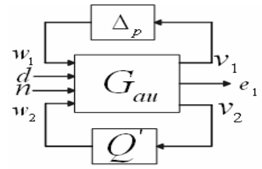


Fig. 4 An equivalent form of the RIC system with the disturbance

In Fig. 4, G_{au} is the generalized plant, Δ_p is the perturbation of the plant, and $Q' = QP_n^{-1}$ is the transfer function to be solved. From Fig. 4, it can be seen that the input and output of G_{au} satisfies:

$$\begin{bmatrix} v_1 \\ e_1 \\ \hline v_2 \end{bmatrix} = G_{au} \begin{bmatrix} w_1 \\ d \\ n \\ \hline w_2 \end{bmatrix} \tag{5}$$

,where G_{au} has 4 inputs w_1, d, n, w_2 and 3 outputs v_1, e_1 and v_2. They correspond to different physical variables respectively, as Fig.5 illustrates.

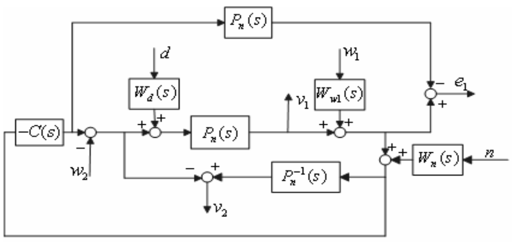


Fig. 5 The structure of generalized plant G_{au}

Note that e_1 is the error between the output of practical plant and output of nominal plant under the same control input u. The ability to reject disturbance and compensate model perturbation can be evaluated by $|e_1|$.

III. THE STABILITY ANALYSISES AND DESIGN OF H_∞ ROBUST DOB

As to satisfy the robust performance mentioned above, the DOB design does not only need to guarantee the internal stability of the system shown in Fig. 4, but also need to restrict the influence from the external disturbance, model perturbation and noise to e_1. To design such a DOB, it is merely needed to solve an equivalent H_∞ sub-optimal problem in practice. If the optimal H_∞ solution can be obtained, a robust DOB optimal in the sense of H_∞ norm would be acquired. In this section, the sufficient condition for robust stability of the closed loop system with DOB is first proposed, after that the procedures to design the H_∞ robust DOB are introduced in detail.

A. Sufficient conditions for robust stability and H_∞ performance of the closed loop system with DOB

To solve Q', first of all, the generalized plant G_{au}. From Fig. 5 we know that:

$$G_{au} = \begin{bmatrix} \dfrac{-CP_nW_{w_1}}{1+CP_n} & \dfrac{P_nW_d}{1+CP_n} & \dfrac{-CP_nW_n}{1+CP_n} & \dfrac{-P_n}{1+CP_n} \\ W_{w_1} & P_nW_d & 0 & -P_n \\ W_{w_1} & P_nW_d & W_n & 0 \end{bmatrix} \tag{6}$$

In the equation above, W_d, $W_n$and W_{w1} are the weighted transfer functions based on the amplitude-frequency characteristics of disturbance d , noise n and model perturbation Δ_p respectively. Generally, disturbance d is regarded as low frequency signal, while noise n high frequency signal [6]. Therefore, W_d is chosen to be a low-pass transfer function while W_n a high-pass transfer function. Note that the amplitude frequency characteristics of W_d, W_n and W_{w1} should cover those of d, n and Δ_p respectively. Because the state-space model of the generalized plant is needed for the LMI method, it should be firstly obtained the minimal realization from the irreducible matrix fraction description of G_{au} [7]. With the minimal realization of G as:

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

The two-port system can be obtained by dividing matrix B, C, and D according to (5) as follows:

$$G_{au} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array}\right] \tag{7}$$

In order to attenuate the difference between the nominal model and the internal loop of the RIC system, which is caused by disturbance, noise and model perturbation, we should keep the norm $\sup_{\|u\|_2 \neq 0}(\|e_1\|_2 / \|[w_1, d, n]^T\|_2)$ under bound. Note that the L_2 induced gain for the system is equivalent to the H_∞ gain [9][10]. From small gain theorem and LMI method, the following theorem is obtained, which gives the sufficient condition to guarantee the internal stability and H_∞ performance.

Theorem 1: Given a positive number γ', assuming that (A,B_2) is stabilizable; (A,C_2) is detectable, the sufficient condition for the existence of DOB, which can guarantee the internal stability of the closed loop system and $\sup_{\|u\|_2 \neq 0}(\|e_1\|_2 / \|[w_1, d, n]^T\|_2) < \gamma'$, is there exits the solution matrix X>0, Y>0 satisfying the following LMIs:

$$\begin{bmatrix} N_o^T & \\ & I \end{bmatrix} \begin{bmatrix} A^T X + XA & XB_1 & C_1^T \\ B_1^T X & -\gamma I & D_{11}^T \\ C_1 & D_{11} & -\gamma I \end{bmatrix} \begin{bmatrix} N_o & \\ & I \end{bmatrix} < 0 \tag{8}$$

$$\begin{bmatrix} N_c^T & \\ & I \end{bmatrix} \begin{bmatrix} AY + YA^T & YC_1^T & B_1 \\ C_1 Y & -\gamma I & D_{11} \\ B_1^T & D_{11}^T & -\gamma I \end{bmatrix} \begin{bmatrix} N_c & \\ & I_2 \end{bmatrix} < 0 \tag{9}$$

$$\begin{bmatrix} X & I \\ I & Y \end{bmatrix} \geq 0 \tag{10}$$

,where γ=min(1/$\|\Delta_p\|$, γ'). In the LMIs, ImN_o=ker[C_2 D_{21}]; ImN_c=ker[B_2^T D_{12}^T].

Proof: If LMI (8), (9), (10) have positive definite solution matrix X and Y, according to [8], then there exists a transfer function Q' satisfying $\|F_l(G_{au}, Q')\|_\infty < \gamma$. From Fig. 4, it can be seen that:

$$F_l(G_{au}, Q') = Y(s)/U(s)$$

, where $Y(s)=[v_1(s),e_1(s)]^T$; $U(s)=[w_1(s),d(s),n(s)]^T$.

From [10], it can be known that,

$\|F_l(G_{au}, Q')\|_\infty = \sup_{U \neq 0}(\|Y(s)\|_2 / \|U(s)\|_2)$

Since $\|[v_1, e_1]^T\|_2 \geq \|e_1\|_2$, we have

$$\sup_{U \neq 0}(\|e_1\|_2 / \|[w_1, d, n]^T\|_2) \leq \|F_l(G_{au}, Q')\|_\infty \leq \gamma \tag{11}$$

Similarly, $\sup_{U \neq 0}(\|v_1\|_2 / \|[w_1, d, n]^T\|_2) \leq \gamma$.

When $w_1 \neq 0$, d=0 and n=0, we have $\|[w_1,d,n]^T\|_2 = \|w_1\|_2$.

Therefore $\sup_{w_1 \neq 0, d=0, n=0}(\|v_1\|_2 / \|[w_1, d, n]^T\|_2) = \sup_{w_1 \neq 0}(\|v_1\|_2 / \|w_1\|_2)$

and $\sup_{w_1 \neq 0, d=0, n=0}(\|v_1\|_2 / \|[w_1, d, n]^T\|_2) \leq \sup_{U \neq 0}(\|v_1\|_2 / \|[w_1, d, n]^T\|_2)$,

So that:

$$\sup_{w_1 \neq 0}(\|v_1\|_2 / \|w_1\|_2) \leq \gamma \tag{12}$$

Then it follows from (11), (12) and γ=min(1/$\|\Delta_p\|$, γ').

that：
$$\begin{cases} \sup_{w_1 \neq 0}(\|v_1\|_2 / \|w_1\|_2) \leq 1/\|\Delta_p\|_\infty \\ \sup_{U \neq 0}(\|e_1\|_2 / \|[w_1, d, n]^T\|_2) \leq \gamma' \end{cases} \tag{13}$$

According to the small gain theorem [10], it can be derived from (13) that the closed loop system as shown in Fig. 4 is internally stable. □

B. The procedure to solve the H_∞ optimal DOB

The procedure to acquire the H_∞ sub-optimal solution Q' is as follows [9]:

Step 1: solve LMI (8)~(9), then obtain positive definite matrixes X, Y;

Step 2: solve the matrix X_2 from $X - Y^{-1} = X_2 X_2^T$, then use matrixes X and X_2 to construct a matrix:

$$X_{cl} = \begin{bmatrix} X & X_2^T \\ X_2 & I \end{bmatrix}.$$

Step 3: submit X_{cl} to the following linear matrix inequality.

$$H_{X_{cl}} + P_{X_{cl}}^T Q' L + L^T Q'^T P_{X_{cl}} < 0 \tag{14}$$

Where $H_{X_{cl}} = \begin{bmatrix} A_0^T X_{cl} + X_{cl} A_0 & X_{cl} B_0 & C_0^T \\ B_0^T X_{cl} & -I & D_{11}^T \\ C_0 & D_{11} & -I \end{bmatrix}$,

$P_{X_{cl}} = [\bar{B}^T X_{cl} \quad 0 \quad \bar{D}_{12}^T], L = [\bar{C} \quad \bar{D}_{21} \quad 0]$

$A_0 = \begin{bmatrix} A & 0 \\ 0 & 0 \end{bmatrix}, B_0 = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, C_0 = [C_1 \quad 0];$

$\bar{B} = \begin{bmatrix} 0 & B_2 \\ I & 0 \end{bmatrix}, \bar{C} = \begin{bmatrix} 0 & I \\ C_2 & 0 \end{bmatrix}, \bar{D}_{12} = [0 \quad D_{12}], \bar{D}_{21} = \begin{bmatrix} 0 \\ D_{21} \end{bmatrix}$

Then the LMI to solve Q' is obtained. Applying the LMI Toolbox of Matlab can solve the parameter matrix of Q'. Furthermore, if the LMI feasibility problem in step1 is replaced by the following LMI optimization problem, the H_∞ robust optimal DOB can be obtained.

$$\min_{X,Y} \gamma$$

$$\textit{subject to}: (8), (9), (10)$$

IV. THE DESIGN AND ANALYSIS OF H_∞ ROBUST OPTIMAL DOB FOR THE TWO-AXIS SERVO TABLE

In this section, the structure and modelling of the two-axis servo table are first introduced. Then a robust DOB is designed with the novel method mentioned above; whereas conventional DOBs are also deduced according to the classic design method. Finally, the simulation results and analysis are provided.

A. Structure and modelling of the two-axis servo table

The two-axis servo table is comprised of two orthogonal rotating axes（horizontal axis and pitching axis）, which are driven by two torque motor respectively, as shown in the following figure.

Fig. 6 Two-axis servo table

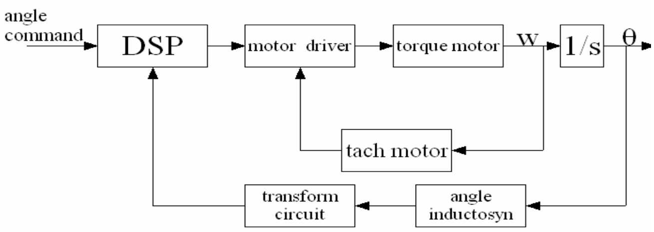


Fig. 7 The block diagram of two-axis servo table

Since there is an unbalanced torque exerted on the pitching axis, during rotation its output speed is affected by disturbance. The inertia of the horizontal axis varies in a large range when the pitching axis rotates, thus the model of horizontal axis has a considerable uncertainty. Furthermore, friction is another source of disturbance. When the servo table rotates, the speed smoothness is deteriorated by disturbance and model perturbation. Therefore, both disturbance as well as model perturbation should be considered during the DOB design. Here, we only take the horizontal axis for example, for the DOB designs of the two axes are similar. The electrical part of the horizontal axis can be described as follows:

$$\begin{cases} T - bv - f_{cs} = J\dot{v} \\ T = K_m u - \tau_m \dot{T} \end{cases} \tag{15}$$

In (15), T denotes output torque of the motor; b denotes the vicious friction coefficient; f_{cs} is the sum of column friction and static friction; J denotes the moment of inertia; K_m is the motor torque coefficient; u denotes the input voltage of the motor; τ_m is the time constant of motor's current. Ignoring the friction f_{cs} in (15), the nominal model of the electrical part of the horizontal axis is:

$$\frac{v(s)}{u(s)} = \frac{K_m}{\tau_m J s^2 + (\tau_m b + J)s + b} \tag{16}$$

After estimating the unknown parameter in (16) from the input-output data of the plant with least square estimation method, the nominal model is obtained as follows:

$$P_n(s) = \frac{v(s)}{u(s)} = \frac{100}{s^2 + 20s + 10} \tag{17}$$

The weighted transfer functions are chosen according to the amplitude-frequency characteristic of disturbance, noise and model perturbation respectively.

$$W_d = \frac{(s+100)^2}{(s+10)^2},\quad W_n = \frac{0.1(s+50)}{s+1000},$$

$$W_{w_1} = \frac{100}{3s^2 + 60s + 10} - \frac{100}{s^2 + 20s + 10}$$

Their frequency characteristics are as follows:

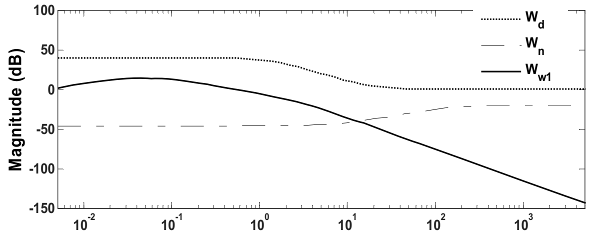


Fig. 8 Frequency characteristics of W_d, W_n and W_{w1}

B. Design of the robust DOB for the two-axis servo table

According to the nominal model (17), the controller is designed as $C(s)$=1+0.5/s. Substituting W_d, W_n, W_{w1}, C and P_n to (6), the parameter of the generalized plant is obtained. Then solving the Q' corresponding to $\min\|F_l(G_{au},Q')\|_\infty$ with the method mentioned above, a low-pass filter Q is acquired. It has the form as follows:

$$Q = P_n Q' = \frac{\begin{array}{l} 2.22\times10^6 s^9 + 2.43\times10^9 s^8 + 2.21\times10^{11} s^7 + 8.8\times10^{12} s^6 \\ +1.9\times10^{14} s^5 + 2.3\times10^{15} s^4 + 1.36\times10^{16} s^3 + 2.6\times10^{16} s^2 \\ +1.67\times10^{16} s + 3.54\times10^{15} \end{array}}{\begin{array}{l} s^{11} + 8543 s^{10} + 1.03\times10^7 s^9 + 3.05\times10^9 s^8 + 2.41\times10^{11} s^7 \\ +9.15\times10^{12} s^6 + 1.94\times10^{14} s^5 + 2.32\times10^{15} s^4 + 1.37\times10^{16} s^3 \\ +2.62\times10^{16} s^2 + 1.71\times10^{16} s + 3.59\times10^{15} \end{array}} \tag{18}$$

From (18), it is obviously that the stable state gain is very close to one, which is identical to conventional Q-filter design method. However, since the 11-th order of (18), it is too high for practical use. Therefore, the order of Q is needed to reduce. After a balanced model reduction, Q' turns out to be a 4-th order transfer function. If the order becomes lower, the frequency characteristic will have considerable change from the original one, which leads to quite a bit change in performance. Finally, a 6-th order Q filter is obtained.

$$Q_{de} = P_n Q'_{de} = \frac{\begin{array}{c} 2.22\times10^6\ s^4 + 2.3\times10^9\ s^3 + 7.79\times10^{10}\ s^2 \\ +6.69\times10^{11} s + 3.59\times10^{11} \end{array}}{\begin{array}{c} s^6 + 8483 s^5 + 9.85\times10^6 s^4 + 2.44\times10^9 s^3 \\ +7.80\times10^{10}\ s^2 + 6.81\times10^{11}\ s + 3.29\times10^{11} \end{array}} \tag{19}$$

Then a comparison between the DOBs designed by the proposed method and the conventional method is made. The simulation shows that the H_∞ robust optimal DOB has better performance than the conventional ones when noise and model perturbation act simultaneously. First, we use the method suggested by T. Umeno, Y. Hori., 1991 to design a second order Q-filter which is denoted as Q_c:

$$Q_c = \frac{3\tau s + 1}{(\tau s)^3 + 3(\tau s)^2 + 3\tau s + 1} \tag{20}$$

,where τ is an adjustment time constant. The DOB responses to disturbance faster if τ is smaller. Nevertheless, it would be more sensitive to noise. On the contrary, the larger the value of τ, the slower the response of DOB to disturbance becomes. However, the anti-noise ability will be enhanced. Fig.9 shows that the frequency characteristics of the Q-filter corresponding to the H_∞ robust optimal DOB (Q), its order reduced form Q_{de} and the conventional Q-filter Q_c.

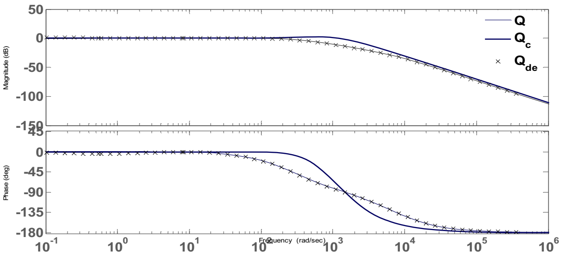


Fig. 9 Frequency characteristics of Q, Q_{de} and Qc

From Fig. 9, it can be seen that the order reduction has little influence to the Q-filter Q.

C. Simulation and analysis

The simulation mentioned below presents that the order reduced DOB still keeps a good performance. During the simulation, a band-limited white noise with power 0.001 is used, which represents the noise in the speed feedback of the tach motor. The disturbance is chosen as a sinusoidal torque with amplitude of 1 $n{\cdot}m$ and frequency of 1 Hz. It's assumed that the load inertia varies in a sinusoidal form with amplitude of 30% of the nominal inertia and frequency of 0.1 Hz. The plant used in the simulation is as follows:

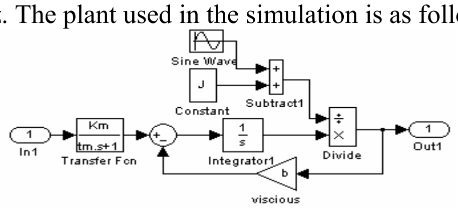
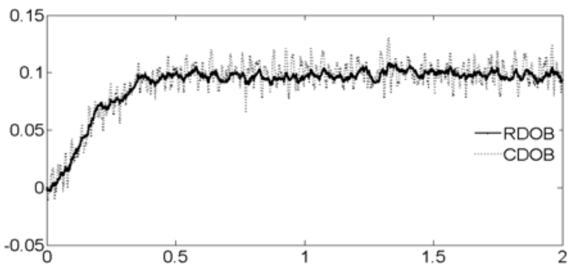
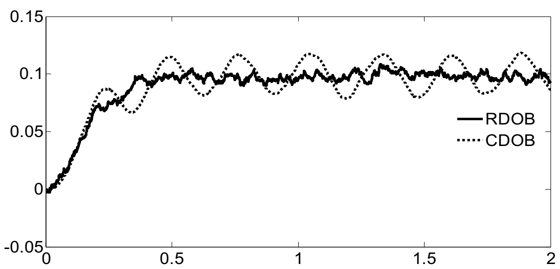


Fig. 10 The simulation model of the plant

The system is affected by a time variant perturbation. Under the influence of noise, the signal to noise ratio of the tach motor feedback is very low in the low speed rotation, which makes it difficult to guarantee the smoothness of low speed motion. In consideration of this, a step signal of small magnitude is used as an input in the simulation. A comparison is made between DOBs designed with proposed method and conventional method respectively in the simulation. The response curve is obtained, in which RDOB denotes the H_∞ robust optimal DOB and CDOB denotes the DOB designed with the conventional method. Different value of τ is chosen during the design of the CDOB. τ is chosen to be 0.001s and 0.015s respectively.



(a) Response of RDOB and CDOB with τ＝0.001s



(b) Response of RDOB and CDOB with τ=0.015s

Fig. 11 Comparison between H_∞ robust DOB and conventional DOB

From Fig. 11, it can be seen that when disturbance, noise and model perturbation are exerted to the system simultaneously, H_∞ robust DOB has better performance than the conventional DOB. Using the H_∞ robust DOB can improve the smoothness of the servo table's low speed motion. Furthermore, the simulation results also show that when the time constant τ is too large, the compensation ability of the conventional DOB would be weaken, then the system is easily affected by disturbance and model perturbation, even become unstable. If the time constant τ is too small, the system will be sensitive to noise. H_∞ robust DOB can trade off pretty well between the disturbance compensation and noise rejection.

V. CONCLUSIONS

A novel design method of H_∞ robust DOB is proposed in this paper. This method takes external disturbance, measurement noise and model perturbation into account simultaneously. The robust DOB designed with the proposed method is optimal in the sense of the H_∞ norm. Furthermore, the design of DOB is practically transformed into a Q-filter design problem. Through introducing weighted transfer functions, the Q-filter design problem can be treated as an H_∞ control problem, and then the LMI method can be used to acquire the solution. The simulation results show that, under the influence of external disturbance, measurement noise and model perturbation, H_∞ robust DOB has better robust performance than the conventional DOB.

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